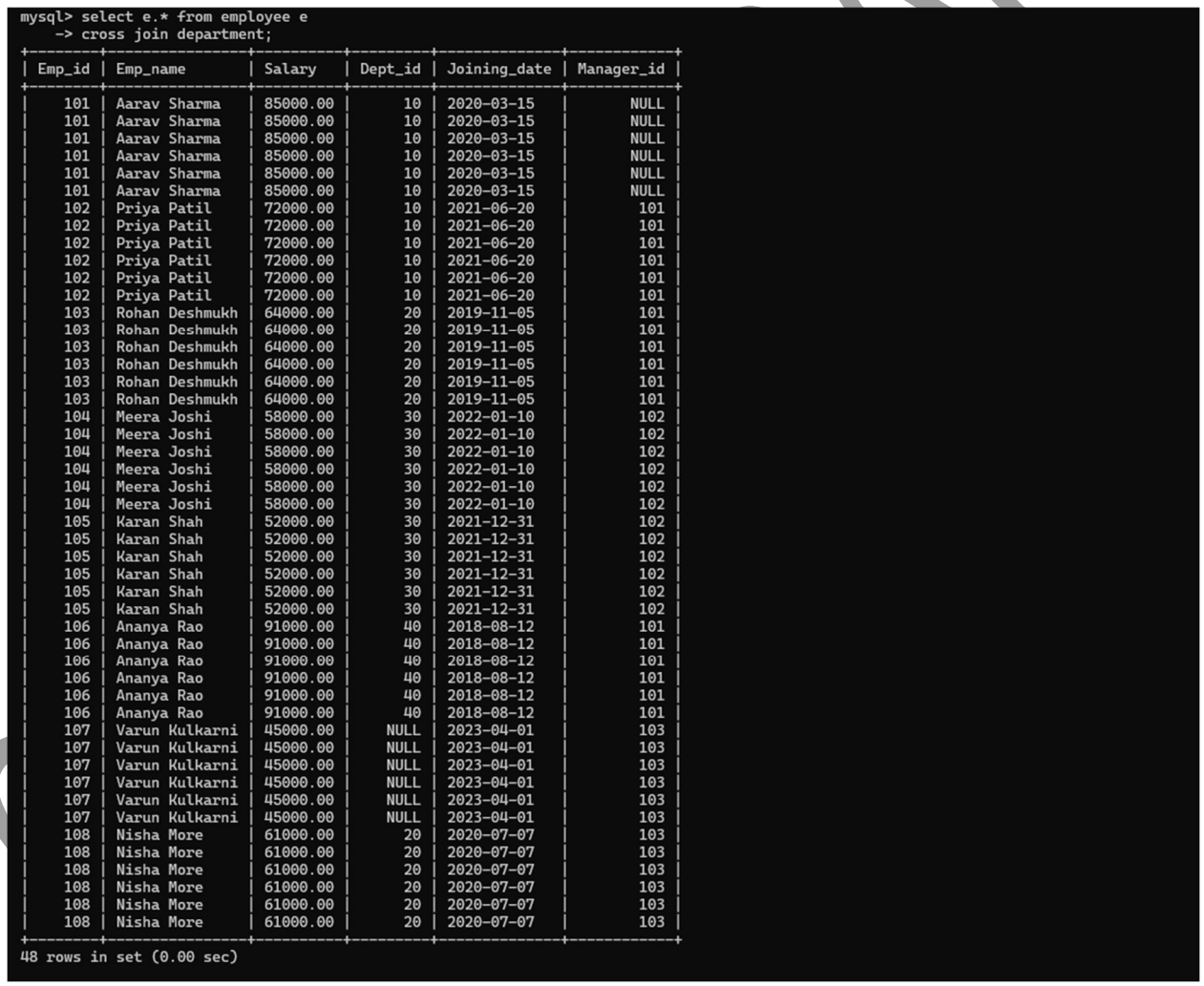
PGCP-AC AUGUST 26
Assignment No- 4

Write Create a Employee ,Department & Project Tables .Add atleast 5 records in each table & Write a following Queries.

1. Write a Relational Algebra expression to perform **Cartesian Product (Cross Join)** between Emp and Dept tables.



```
mysql> select e.* from employee e
    -> cross join department;
+--------+---------------+----------+---------+--------------+------------+
| Emp_id | Emp_name      | Salary   | Dept_id | Joining_date | Manager_id |
+--------+---------------+----------+---------+--------------+------------+
|    101 | Aarav Sharma  | 85000.00 |      10 | 2020-03-15   |       NULL |
|    101 | Aarav Sharma  | 85000.00 |      10 | 2020-03-15   |       NULL |
|    101 | Aarav Sharma  | 85000.00 |      10 | 2020-03-15   |       NULL |
|    101 | Aarav Sharma  | 85000.00 |      10 | 2020-03-15   |       NULL |
|    101 | Aarav Sharma  | 85000.00 |      10 | 2020-03-15   |       NULL |
|    101 | Aarav Sharma  | 85000.00 |      10 | 2020-03-15   |       NULL |
|    102 | Priya Patil   | 72000.00 |      10 | 2021-06-20   |        101 |
|    102 | Priya Patil   | 72000.00 |      10 | 2021-06-20   |        101 |
|    102 | Priya Patil   | 72000.00 |      10 | 2021-06-20   |        101 |
|    102 | Priya Patil   | 72000.00 |      10 | 2021-06-20   |        101 |
|    102 | Priya Patil   | 72000.00 |      10 | 2021-06-20   |        101 |
|    102 | Priya Patil   | 72000.00 |      10 | 2021-06-20   |        101 |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |
|    104 | Meera Joshi   | 58000.00 |      30 | 2022-01-10   |        102 |
|    104 | Meera Joshi   | 58000.00 |      30 | 2022-01-10   |        102 |
|    104 | Meera Joshi   | 58000.00 |      30 | 2022-01-10   |        102 |
|    104 | Meera Joshi   | 58000.00 |      30 | 2022-01-10   |        102 |
|    104 | Meera Joshi   | 58000.00 |      30 | 2022-01-10   |        102 |
|    104 | Meera Joshi   | 58000.00 |      30 | 2022-01-10   |        102 |
|    105 | Karan Shah    | 52000.00 |      30 | 2021-12-31   |        102 |
|    105 | Karan Shah    | 52000.00 |      30 | 2021-12-31   |        102 |
|    105 | Karan Shah    | 52000.00 |      30 | 2021-12-31   |        102 |
|    105 | Karan Shah    | 52000.00 |      30 | 2021-12-31   |        102 |
|    105 | Karan Shah    | 52000.00 |      30 | 2021-12-31   |        102 |
|    105 | Karan Shah    | 52000.00 |      30 | 2021-12-31   |        102 |
|    106 | Ananya Rao    | 91000.00 |      40 | 2018-08-12   |        101 |
|    106 | Ananya Rao    | 91000.00 |      40 | 2018-08-12   |        101 |
|    106 | Ananya Rao    | 91000.00 |      40 | 2018-08-12   |        101 |
|    106 | Ananya Rao    | 91000.00 |      40 | 2018-08-12   |        101 |
|    106 | Ananya Rao    | 91000.00 |      40 | 2018-08-12   |        101 |
|    106 | Ananya Rao    | 91000.00 |      40 | 2018-08-12   |        101 |
|    107 | Varun Kulkarni | 45000.00 |    NULL | 2023-04-01   |        103 |
|    107 | Varun Kulkarni | 45000.00 |    NULL | 2023-04-01   |        103 |
|    107 | Varun Kulkarni | 45000.00 |    NULL | 2023-04-01   |        103 |
|    107 | Varun Kulkarni | 45000.00 |    NULL | 2023-04-01   |        103 |
|    107 | Varun Kulkarni | 45000.00 |    NULL | 2023-04-01   |        103 |
|    107 | Varun Kulkarni | 45000.00 |    NULL | 2023-04-01   |        103 |
|    108 | Nisha More    | 61000.00 |      20 | 2020-07-07   |        103 |
|    108 | Nisha More    | 61000.00 |      20 | 2020-07-07   |        103 |
|    108 | Nisha More    | 61000.00 |      20 | 2020-07-07   |        103 |
|    108 | Nisha More    | 61000.00 |      20 | 2020-07-07   |        103 |
|    108 | Nisha More    | 61000.00 |      20 | 2020-07-07   |        103 |
|    108 | Nisha More    | 61000.00 |      20 | 2020-07-07   |        103 |
+--------+---------------+----------+---------+--------------+------------+
48 rows in set (0.00 sec)
```

2. Write a SQL query to perform an **INNER JOIN** between Emp and Dept tables displaying EmpName and DeptName.

```
mysql> select e.emp_name,d.dept_name from employee e
    -> inner join department d
    -> on e.dept_id = d.dept_id;
+----------------+-----------+
| emp_name       | dept_name |
+----------------+-----------+
| Aarav Sharma   | IT        |
| Priya Patil    | IT        |
| Rohan Deshmukh | HR        |
| Nisha More     | HR        |
| Meera Joshi    | Sales     |
| Karan Shah     | Sales     |
| Ananya Rao     | Finance   |
+----------------+-----------+
7 rows in set (0.00 sec)
```

3. Write a SQL query to perform a **LEFT OUTER JOIN** between Emp and Dept tables to display all employees along with their department information, including employees without a department.

```
mysql> select e.*,d.* from employee e
    -> left join department d
    -> on e.dept_id = d.dept_id;
+--------+----------------+----------+---------+--------------+------------+---------+-----------+--------+
| Emp_id | Emp_name       | Salary   | Dept_id | Joining_date | Manager_id | Dept_id | Dept_name | city   |
+--------+----------------+----------+---------+--------------+------------+---------+-----------+--------+
|    101 | Aarav Sharma   | 85000.00 |      10 | 2020-03-15   |       NULL |      10 | IT        | Mumbai |
|    102 | Priya Patil    | 72000.00 |      10 | 2021-06-20   |        101 |      10 | IT        | Mumbai |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |      20 | HR        | Pune   |
|    104 | Meera Joshi    | 58000.00 |      30 | 2022-01-10   |        102 |      30 | Sales     | Mumbai |
|    105 | Karan Shah     | 52000.00 |      30 | 2021-12-31   |        102 |      30 | Sales     | Mumbai |
|    106 | Ananya Rao     | 91000.00 |      40 | 2018-08-12   |        101 |      40 | Finance   | Delhi  |
|    107 | Varun Kulkarni | 45000.00 |    NULL | 2023-04-01   |        103 |    NULL | NULL      | NULL   |
|    108 | Nisha More     | 61000.00 |      20 | 2020-07-07   |        103 |      20 | HR        | Pune   |
+--------+----------------+----------+---------+--------------+------------+---------+-----------+--------+
8 rows in set (0.00 sec)
```

4. Write a SQL query to perform a **RIGHT OUTER JOIN** between Emp and Dept tables.

```
mysql> select e.*,d.* from employee e
    -> right join department d
    -> on e.dept_id = d.dept_id;
+--------+----------------+----------+---------+--------------+------------+---------+------------+--------+
| Emp_id | Emp_name       | Salary   | Dept_id | Joining_date | Manager_id | Dept_id | Dept_name  | city   |
+--------+----------------+----------+---------+--------------+------------+---------+------------+--------+
|    101 | Aarav Sharma   | 85000.00 |      10 | 2020-03-15   |       NULL |      10 | IT         | Mumbai |
|    102 | Priya Patil    | 72000.00 |      10 | 2021-06-20   |        101 |      10 | IT         | Mumbai |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |      20 | HR         | Pune   |
|    108 | Nisha More     | 61000.00 |      20 | 2020-07-07   |        103 |      20 | HR         | Pune   |
|    104 | Meera Joshi    | 58000.00 |      30 | 2022-01-10   |        102 |      30 | Sales      | Mumbai |
|    105 | Karan Shah     | 52000.00 |      30 | 2021-12-31   |        102 |      30 | Sales      | Mumbai |
|    106 | Ananya Rao     | 91000.00 |      40 | 2018-08-12   |        101 |      40 | Finance    | Delhi  |
|   NULL | NULL           |     NULL |    NULL | NULL         |       NULL |      50 | Marketing  | Nagpur |
|   NULL | NULL           |     NULL |    NULL | NULL         |       NULL |      60 | Operations | Pune   |
+--------+----------------+----------+---------+--------------+------------+---------+------------+--------+
9 rows in set (0.00 sec)
```

5. Write a SQL query to perform a **NATURAL JOIN** between Emp and Dept.

```
mysql> select e.*,d.* from employee e
    -> natural join department d;
+--------+---------------+---------+---------+--------------+------------+---------+-----------+--------+
| Emp_id | Emp_name      | Salary  | Dept_id | Joining_date | Manager_id | Dept_id | Dept_name | city   |
+--------+---------------+---------+---------+--------------+------------+---------+-----------+--------+
|    101 | Aarav Sharma  | 85000.00 |      10 | 2020-03-15   |       NULL |      10 | IT        | Mumbai |
|    102 | Priya Patil   | 72000.00 |      10 | 2021-06-20   |        101 |      10 | IT        | Mumbai |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |      20 | HR        | Pune   |
|    108 | Nisha More    | 61000.00 |      20 | 2020-07-07   |        103 |      20 | HR        | Pune   |
|    104 | Meera Joshi   | 58000.00 |      30 | 2022-01-10   |        102 |      30 | Sales     | Mumbai |
|    105 | Karan Shah    | 52000.00 |      30 | 2021-12-31   |        102 |      30 | Sales     | Mumbai |
|    106 | Ananya Rao    | 91000.00 |      40 | 2018-08-12   |        101 |      40 | Finance   | Delhi  |
+--------+---------------+---------+---------+--------------+------------+---------+-----------+--------+
7 rows in set (0.00 sec)
```

6. Write a SQL query to perform a **CROSS JOIN** between Emp and Project table.

```
mysql> select emp_id,emp_name,p.project_name from employee
    -> cross join project p;
+--------+---------------+---------------------+
| emp_id | emp_name      | project_name        |
+--------+---------------+---------------------+
|    101 | Aarav Sharma  | Process Improvement |
|    101 | Aarav Sharma  | Brand Campaign      |
|    101 | Aarav Sharma  | Audit Automation    |
|    101 | Aarav Sharma  | Sales Analytics     |
|    101 | Aarav Sharma  | Recruitment System  |
|    101 | Aarav Sharma  | Customer Portal     |
|    102 | Priya Patil   | Process Improvement |
|    102 | Priya Patil   | Brand Campaign      |
|    102 | Priya Patil   | Audit Automation    |
|    102 | Priya Patil   | Sales Analytics     |
|    102 | Priya Patil   | Recruitment System  |
|    102 | Priya Patil   | Customer Portal     |
|    103 | Rohan Deshmukh | Process Improvement |
|    103 | Rohan Deshmukh | Brand Campaign      |
|    103 | Rohan Deshmukh | Audit Automation    |
|    103 | Rohan Deshmukh | Sales Analytics     |
|    103 | Rohan Deshmukh | Recruitment System  |
|    103 | Rohan Deshmukh | Customer Portal     |
|    104 | Meera Joshi   | Process Improvement |
|    104 | Meera Joshi   | Brand Campaign      |
|    104 | Meera Joshi   | Audit Automation    |
|    104 | Meera Joshi   | Sales Analytics     |
|    104 | Meera Joshi   | Recruitment System  |
|    104 | Meera Joshi   | Customer Portal     |
|    105 | Karan Shah    | Process Improvement |
|    105 | Karan Shah    | Brand Campaign      |
|    105 | Karan Shah    | Audit Automation    |
|    105 | Karan Shah    | Sales Analytics     |
|    105 | Karan Shah    | Recruitment System  |
|    105 | Karan Shah    | Customer Portal     |
|    106 | Ananya Rao    | Process Improvement |
|    106 | Ananya Rao    | Brand Campaign      |
|    106 | Ananya Rao    | Audit Automation    |
|    106 | Ananya Rao    | Sales Analytics     |
|    106 | Ananya Rao    | Recruitment System  |
|    106 | Ananya Rao    | Customer Portal     |
|    107 | Varun Kulkarni | Process Improvement |
|    107 | Varun Kulkarni | Brand Campaign      |
|    107 | Varun Kulkarni | Audit Automation    |
|    107 | Varun Kulkarni | Sales Analytics     |
|    107 | Varun Kulkarni | Recruitment System  |
|    107 | Varun Kulkarni | Customer Portal     |
|    108 | Nisha More    | Process Improvement |
|    108 | Nisha More    | Brand Campaign      |
|    108 | Nisha More    | Audit Automation    |
|    108 | Nisha More    | Sales Analytics     |
|    108 | Nisha More    | Recruitment System  |
|    108 | Nisha More    | Customer Portal     |
+--------+---------------+---------------------+
48 rows in set (0.00 sec)
```

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7. Write a SQL query to **create a new table EmpBackup with the same structure as Emp but no data**.

```
mysql> create table EmpBackup like employee;
Query OK, 0 rows affected (0.05 sec)

mysql> desc empBackup;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| Emp_id       | int           | NO   | PRI | NULL    | auto_increment |
| Emp_name     | varchar(60)   | NO   |     | NULL    |                |
| Salary       | decimal(10,2) | YES  |     | NULL    |                |
| Dept_id      | int           | YES  | MUL | NULL    |                |
| Joining_date | date          | YES  |     | NULL    |                |
| Manager_id   | int           | YES  | MUL | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
6 rows in set (0.01 sec)

mysql> select * from empBackup;
Empty set (0.00 sec)
```

8. Write a SQL query to **copy all data from Emp into EmpBackup table**.

```
mysql> insert into empBackup select * from employee;
Query OK, 8 rows affected (0.02 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> select * from empBackup;
+--------+----------------+----------+---------+--------------+------------+
| Emp_id | Emp_name       | Salary   | Dept_id | Joining_date | Manager_id |
+--------+----------------+----------+---------+--------------+------------+
|    101 | Aarav Sharma   | 85000.00 |      10 | 2020-03-15   |       NULL |
|    102 | Priya Patil    | 72000.00 |      10 | 2021-06-20   |        101 |
|    103 | Rohan Deshmukh | 64000.00 |      20 | 2019-11-05   |        101 |
|    104 | Meera Joshi    | 58000.00 |      30 | 2022-01-10   |        102 |
|    105 | Karan Shah     | 52000.00 |      30 | 2021-12-31   |        102 |
|    106 | Ananya Rao     | 91000.00 |      40 | 2018-08-12   |        101 |
|    107 | Varun Kulkarni | 45000.00 |    NULL | 2023-04-01   |        103 |
|    108 | Nisha More     | 61000.00 |      20 | 2020-07-07   |        103 |
+--------+----------------+----------+---------+--------------+------------+
8 rows in set (0.00 sec)
```

9. Write a SQL query to **create a new table ProjectArchive with the same structure and data as Project**.

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```
mysql> create table projectArchive like project;
Query OK, 0 rows affected (0.05 sec)

mysql> desc projectArchive;
+--------------+---------------+------+-----+---------+-------+
| Field        | Type          | Null | Key | Default | Extra |
+--------------+---------------+------+-----+---------+-------+
| Project_id   | int           | NO   | PRI | NULL    |       |
| Project_name | varchar(80)   | NO   |     | NULL    |       |
| Dept_id      | int           | YES  | MUL | NULL    |       |
| Budget       | decimal(12,2) | YES  |     | NULL    |       |
| Start_date   | date          | YES  |     | NULL    |       |
+--------------+---------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> insert into projectArchive select * from project;
Query OK, 6 rows affected (0.02 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> select * from projectArchive;
+------------+---------------------+---------+-----------+------------+
| Project_id | Project_name        | Dept_id | Budget    | Start_date |
+------------+---------------------+---------+-----------+------------+
|        201 | Customer Portal     |      10 | 450000.00 | 2025-01-15 |
|        202 | Recruitment System  |      20 | 180000.00 | 2025-03-01 |
|        203 | Sales Analytics     |      30 | 320000.00 | 2025-02-10 |
|        204 | Audit Automation    |      40 | 250000.00 | 2025-04-20 |
|        205 | Brand Campaign      |      50 | 150000.00 | 2025-05-05 |
|        206 | Process Improvement |      60 | 210000.00 | 2025-06-12 |
+------------+---------------------+---------+-----------+------------+
6 rows in set (0.00 sec)
```

10. Write a SQL query to create an **AUTO_INCREMENT sequence for EmpID in the Emp table** during table creation.

```
mysql> Create table Employee (
    ->          Emp_id int primary key auto_increment,
    ->          Emp_name varchar(30) not null,
    ->          Salary decimal(10,2),
    ->          Dept_id int not null,
    ->          Joining_date date,
    ->          Manager_id int not null,
    -> constraint fk_emp_dept foreign key (Dept_id) references Department(Dept_id),
    ->         constraint fk_emp_manager foreign key (Manager_id) references Employee(Emp_id)
    ->      );
Query OK, 0 rows affected (0.06 sec)

mysql> desc employee;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| Emp_id       | int           | NO   | PRI | NULL    | auto_increment |
| Emp_name     | varchar(30)   | NO   |     | NULL    |                |
| Salary       | decimal(10,2) | YES  |     | NULL    |                |
| Dept_id      | int           | NO   | MUL | NULL    |                |
| Joining_date | date          | YES  |     | NULL    |                |
| Manager_id   | int           | NO   | MUL | NULL    |                |
+--------------+---------------+------+-----+---------+----------------+
6 rows in set (0.01 sec)
```

11. Write a SQL query using a **subquery** to find all employees whose salary is greater than the **average salary of all employees**.

```
mysql> select emp_id, emp_name , salary from employee where salary > (select avg(salary) from employee);
+--------+--------------+----------+
| emp_id | emp_name     | salary   |
+--------+--------------+----------+
|    101 | Aarav Sharma | 85000.00 |
|    102 | Priya Patil  | 72000.00 |
|    106 | Ananya Rao   | 91000.00 |
+--------+--------------+----------+
3 rows in set (0.00 sec)
```

12. Write a SQL query using a **subquery in WHERE clause** to find employees working in departments located in 'Mumbai'.

```
mysql> select* from employee where dept_id in (select dept_id from department where city = 'Mumbai');
+--------+--------------+----------+---------+--------------+------------+
| Emp_id | Emp_name     | Salary   | Dept_id | Joining_date | Manager_id |
+--------+--------------+----------+---------+--------------+------------+
|    101 | Aarav Sharma | 85000.00 |      10 | 2020-03-15   |       NULL |
|    102 | Priya Patil  | 72000.00 |      10 | 2021-06-20   |        101 |
|    104 | Meera Joshi  | 58000.00 |      30 | 2022-01-10   |        102 |
|    105 | Karan Shah   | 52000.00 |      30 | 2021-12-31   |        102 |
+--------+--------------+----------+---------+--------------+------------+
4 rows in set (0.00 sec)
```

13. Write a SQL query to display all departments where the number of employees is greater than **5**, using a subquery.

```
mysql> select * from department d where (select count(*) from employee e where e.dept_id = d.dept_id)  > 5;
Empty set (0.00 sec)

mysql> select * from department d where (select count(*) from employee e where e.dept_id = d.dept_id)  > 1;
+---------+-----------+--------+
| Dept_id | Dept_name | city   |
+---------+-----------+--------+
|      10 | IT        | Mumbai |
|      20 | HR        | Pune   |
|      30 | Sales     | Mumbai |
+---------+-----------+--------+
3 rows in set (0.00 sec)
```

14. Write a subqueries for find out Max salary of employee.

```
mysql> select * from employee where salary = (select max(salary) from employee);
+--------+-----------+----------+---------+--------------+------------+
| Emp_id | Emp_name  | Salary   | Dept_id | Joining_date | Manager_id |
+--------+-----------+----------+---------+--------------+------------+
|    106 | Ananya Rao | 91000.00 |      40 | 2018-08-12   |        101 |
+--------+-----------+----------+---------+--------------+------------+
1 row in set (0.00 sec)
```

15. Write a self join query for each table.

```
mysql> select e.emp_name as employee,
    -> m.emp_name as manager
    -> from  employee e
    -> left join employee m
    -> on e.manager_id = m.emp_id;
+----------------+----------------+
| employee       | manager        |
+----------------+----------------+
| Aarav Sharma   | NULL           |
| Priya Patil    | Aarav Sharma   |
| Rohan Deshmukh | Aarav Sharma   |
| Meera Joshi    | Priya Patil    |
| Karan Shah     | Priya Patil    |
| Ananya Rao     | Aarav Sharma   |
| Varun Kulkarni | Rohan Deshmukh |
| Nisha More     | Rohan Deshmukh |
+----------------+----------------+
8 rows in set (0.00 sec)
```

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